



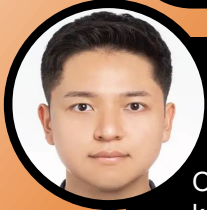
# Weekly Seminar Series

Fridays @ 11AM ET

[https://bit.ly/CHCI\\_VT\\_Weekly\\_Seminar](https://bit.ly/CHCI_VT_Weekly_Seminar)



**Friday, March 27, 2026 @ 11AM ET**



## *CHOIR: Chatbot-mediated Organizational Memory Leveraging Communication in Research Labs*

**Sangwook Lee, Virginia Tech**

Organizations increasingly rely on chat-based platforms for everyday collaboration, where valuable knowledge often emerges but is easily lost in message streams. We present CHOIR, an LLM-based chatbot that supports organizational memory through document-grounded Q&A, Q&A sharing, knowledge extraction from conversations, and AI-assisted document updates. A one-month deployment in multiple organizations reveals tensions between privacy, awareness, and documentation, informing the design of AI-mediated organizational memory systems.

Sangwook Lee is a Ph.D. student in Computer Science at Virginia Tech. He received his M.S. from KAIST and B.S. from Pohang University of Science and Technology in South Korea. His doctoral research aims to bridge technological and information gaps between people through AI-mediated systems.

## *Possible at Last: Using Large Multimodal Models to Evaluate and Secure Mixed-Reality Experiences*

**Maria Gorlatova, Duke University**

When deployed in unconstrained real-world environments, modern mixed reality (MR) experiences can fail in subtle but consequential ways: virtual objects may drift or occlude critical real-world elements; spatial audio cues may contradict visual content; MR-based guidance may suggest actions that are impossible, unsafe, or societally unacceptable. Such highly context-dependent failures, where MR content that is well-suited to one environment can be unsafe in another, cannot be detected by classical signal processing or classical ML approaches alone.

Recent large multimodal models (LMMs) provide a new mechanism for monitoring immersive systems. By jointly reasoning over video streams captured by MR devices, rendered virtual content, spatial audio, and scene context, these models have the potential to identify misaligned overlays, unsafe occlusions, inconsistent audio-visual cues, and policy violations in running MR applications. In this talk I present an LMM-based MR evaluation pipeline that employs LMMs and a collection of auxiliary ML models to analyze continuous multimodal streams and trigger corrective actions when problematic virtual content is detected. A key challenge in LMM-based XR evaluation is reducing inference latency to support near real-time operation. I present a collection of system design choices and lay out a research agenda with the potential to bring down the cost and latency of these systems.

Dr. Maria Gorlatova is the James N. & Elizabeth H. Barton Associate Professor of Electrical and Computer Engineering at Duke University. Her research centers on pervasive computing technologies, with a focus on spatial computing and next-generation augmented reality. She received her Ph.D. in Electrical Engineering from Columbia University and her M.Sc. and B.Sc. degrees (Summa Cum Laude) in Electrical Engineering from the University of Ottawa, Canada. Dr. Gorlatova was named one of the N2Women Rising Stars and has received multiple awards, including the NSF CAREER Award, the DARPA Young Faculty Award (YFA), the DARPA Director's Fellowship, the IEEE ISMAR 2025 Best Paper Award, and research awards from Meta, Cisco, and IBM. She served as the General Co-Chair of the 2025 NSF Workshop on Networking and Systems Challenges in Immersive Computing, and the TPC Co-chair of ACM/IEEE SenSys 2025. For more information, visit <https://gorlatova.pratt.duke.edu/>.

